UNIVERSITY COLLEGE OF ENGINEERING

End Semester Examination

Computer Organization and Architecture (BCS302)

Roll No. _____

Subject: Computer Organization and Architecture

Time: 3 Hours

Max. Marks: 84

Code: BCS302

General Instructions:

1. Attempt all questions.
2. Each question carries marks as indicated in its respective section.
3. Assume suitable data if necessary.

SECTION A — SECTION A

Short Answer Questions

2 marks each | Attempt all questions

Q.No.	Question	Marks	CO
1	A system uses a 16-bit address bus. If the effective address is calculated using a base register and a displacement, analyze how this mechanism supports program relocation in memory.	**2**	**CO1**
2	A specialized processor uses a non-standard floating-point format with a 4-bit exponent and 3-bit mantissa. Determine the binary representation for the decimal value 1.5 in this format.	**2**	**CO2**
3	In a micro-programmed control unit, the control memory consists of 1024 words. Calculate the minimum number of bits required for the Control Address Register (CAR).	**2**	**CO3**
4	A cache memory system has a hit ratio of 0.95. If the cache access time is 10ns and the main memory access time is 100ns, calculate the average memory access time (AMAT).	**2**	**CO4**
5	Evaluate a scenario where multiple I/O devices request service simultaneously. Explain which hardware mechanism ensures that the device with the highest priority is serviced first.	**2**	**CO5**
6	Compare centralized and distributed bus arbitration for a system with 5 master devices in terms of susceptibility to a single point of failure.	**2**	**CO1**
7	Analyze why a Look-Ahead Carry Adder is preferred over a Ripple Carry Adder for high-speed arithmetic, specifically focusing on how it handles carry propagation delay.	**2**	**CO2**

SECTION B — SECTION B

Attempt only 3

7 marks each | Attempt all questions

Q.No.	Question	Marks	CO
8	Analyze a 4-stage pipeline (Fetch, Decode, Execute, Write-back) processing a loop. If a branch instruction at the end of the loop causes a 2-cycle stall every iteration, calculate the total cycles required and the speedup achieved over a non-pipelined system for 100 iterations.	**7**	**CO3**
9	A system implements a 2-level paging scheme with a 32-bit virtual address and a 4KB page size. If each page table entry is 4 bytes, evaluate why this hierarchical structure is more memory-efficient than a single-level page table for a process using only 8MB of its address space.	**7**	**CO4**
10	A high-speed network interface needs to transfer 100MB of data to RAM. Contrast the CPU overhead and total system throughput if the system uses Interrupt-initiated I/O versus Direct Memory Access (DMA). Justify which is better for this specific workload.	**7**	**CO5**
11	You are designing a processor to evaluate complex mathematical expressions (e.g., A*B + C/D). Justify the choice between a Stack-organized CPU and a General Register-organized CPU, focusing on the number of memory accesses required for instruction fetching and data movement.	**7**	**CO1**
12	Perform the multiplication of the signed numbers (-7) and (+5) using Booth's Algorithm. Show the step-by-step register transitions (A, Q, Q-1) and explain why Booth's algorithm is particularly efficient when the multiplier contains long strings of 1s.	**7**	**CO2**

SECTION C — SECTION C

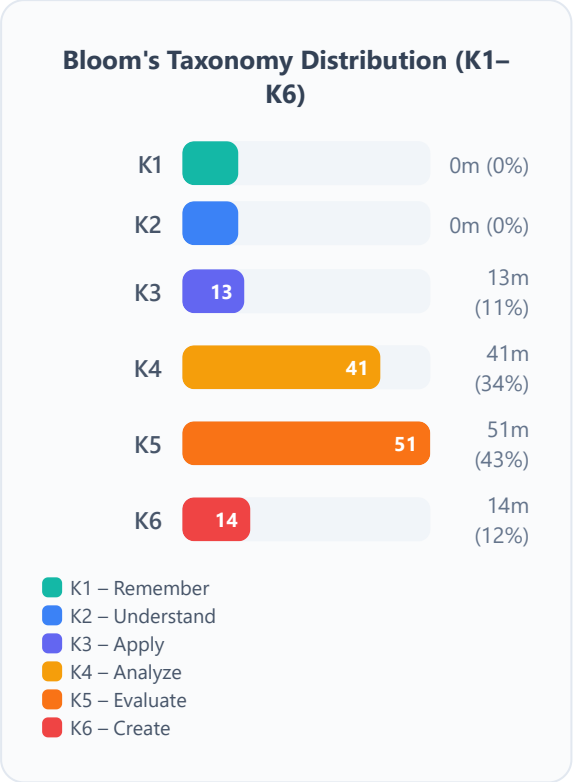
Analytical Choice Questions

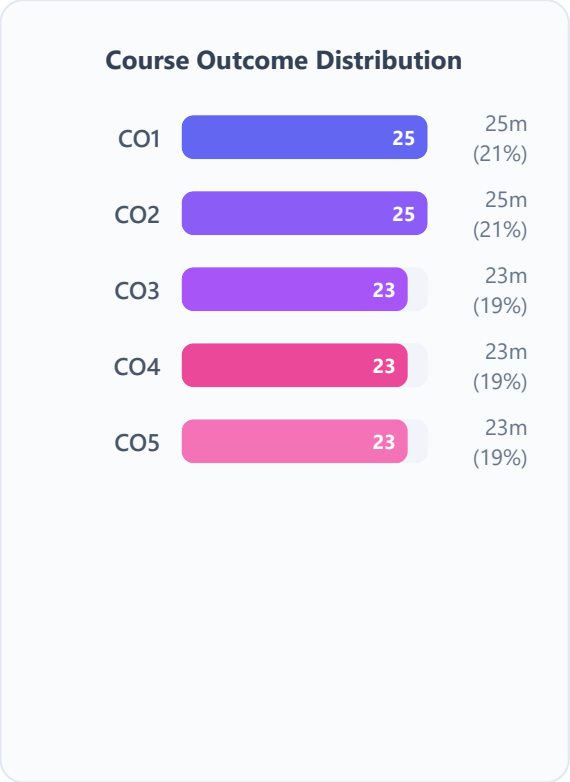
7 marks each | Attempt all questions

Q.No.	Question	Marks	CO
13	**(a)** A multi-core processor shares a single system bus. Design an arbitration logic that prevents device starvation while maintaining high bus throughput during heavy traffic. Compare your design with a simple Fixed Priority scheme.	**7**	**CO1**
	(b) Compare the performance of a processor using 'Register Indirect' addressing versus 'Indexed' addressing for traversing a large 2D matrix stored in row-major order. Which mode reduces the instruction count in the inner loop?	**7**	**CO1**
14	**(a)** A scientific simulation requires summing millions of small floating-point numbers. Analyze how the IEEE 754 precision constraints might lead to significant errors if the sum grows much larger than the individual terms. Propose a mitigation strategy.	**7**	**CO2**
	(b) Design the logical data path for an ALU that can perform both signed multiplication and division. Highlight which registers are reused for both operations and how the control signals differ between the two.	**7**	**CO2**

Q.No.	Question	Marks	CO
15	**(a)** A designer is deciding between Hardwired and Microprogrammed control for a new RISC architecture. Evaluate the trade-offs in terms of instruction execution latency and the complexity involved in adding a new 'Multiply-Accumulate' instruction later.	**7**	**CO3**
	(b) Analyze the impact of 'Read-After-Write' (RAW) data hazards on a 5-stage pipeline. Propose a hardware-based solution such as 'Operand Forwarding' and demonstrate how it reduces pipeline stalls using a sample code sequence.	**7**	**CO3**
16	**(a)** A database application exhibits high spatial locality but very poor temporal locality. Evaluate whether a Direct Mapped Cache or a 4-Way Set-Associative Cache would be more effective for this specific workload, considering the risk of conflict misses.	**7**	**CO4**
	(b) Explain the implementation of Virtual Memory using a Translation Lookaside Buffer (TLB). If the TLB hit rate is 99% and a TLB miss requires a page table walk taking 50ns, analyze the impact on the effective memory access time.	**7**	**CO4**
17	**(a)** A real-time embedded system must process sensor data every 500 microseconds. Compare the suitability of Daisy Chaining versus Vectored Interrupts for this application, focusing on interrupt latency and hardware complexity.	**7**	**CO5**
	(b) Contrast Synchronous and Asynchronous serial communication for a system where the sender and receiver are located 100 meters apart and cannot share a common clock signal. Detail how data synchronization is achieved in the asynchronous case.	**7**	**CO5**

Question Paper Analysis





Bloom's Taxonomy Summary		
Level	**Marks**	**%**
K1	0	0%
K2	0	0%
K3	13	11%
K4	41	34%
K5	51	43%
K6	14	12%

Course Outcome Summary		
CO	**Marks**	**%**
CO1	25	21%
CO2	25	21%
CO3	23	19%
CO4	23	19%
CO5	23	19%